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**WildEcoWatch : An Integrated IoT and AI Framework for
Wildlife Surveillance and Conservation**

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Introduction

- Wildlife conservation is crucial for maintaining biodiversity, but traditional methods like manual patrols and basic camera traps are inefficient and slow.
- The WildEcoWatch project integrates IoT, AI, and machine learning to create a smart, real-time surveillance system.
- It uses IoT devices like cameras and Raspberry Pi to detect intrusions and wildlife disturbances, providing instant alerts to authorities.
- This project bridges technology and forest protection, offering an affordable, scalable solution for wildlife conservation.

Problem Statement

- Wildlife crime, such as poaching and illegal logging, threatens biodiversity, but current surveillance systems are inadequate.
- Traditional methods lack real-time monitoring and instant alerts, leaving wildlife vulnerable.
- The WildEcoWatch project addresses this by leveraging IoT and AI to create an intelligent system that detects intrusions in real-time, ensuring timely responses to protect endangered species and ecosystems.

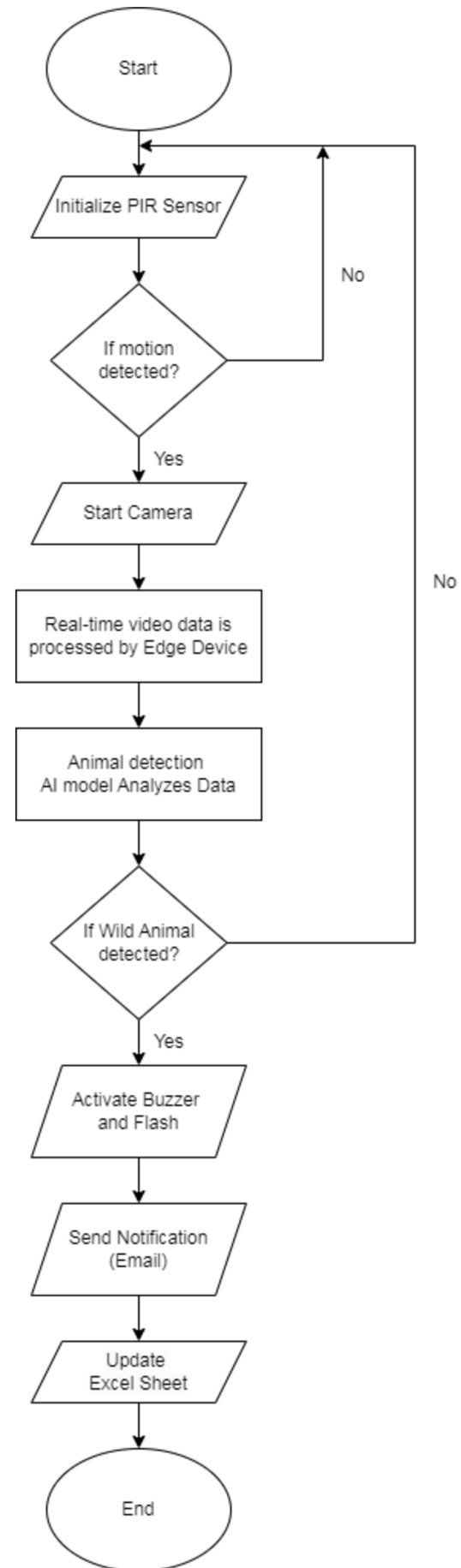
Objectives

1. **Develop an Intelligent Surveillance System:** Create a system using IoT and AI to detect poaching and disruptions in rainforests, utilizing Raspberry Pi.
2. **Real-Time Monitoring and Threat Detection:** Implement IoT sensors and cameras for continuous monitoring, using machine learning to detect threats in real-time.
3. **Instant Alerts and Notifications:** Provide real-time alerts via SMS, emails, notifications, or voice alerts to ensure timely intervention.
4. **Scalability and Sustainability:** Design a scalable, cost-effective system that operates in remote areas using solar power and energy-efficient components.
5. **Enhance Wildlife Conservation Efforts:** Contribute to global conservation by reducing human-wildlife conflicts and protecting endangered species through advanced technology.

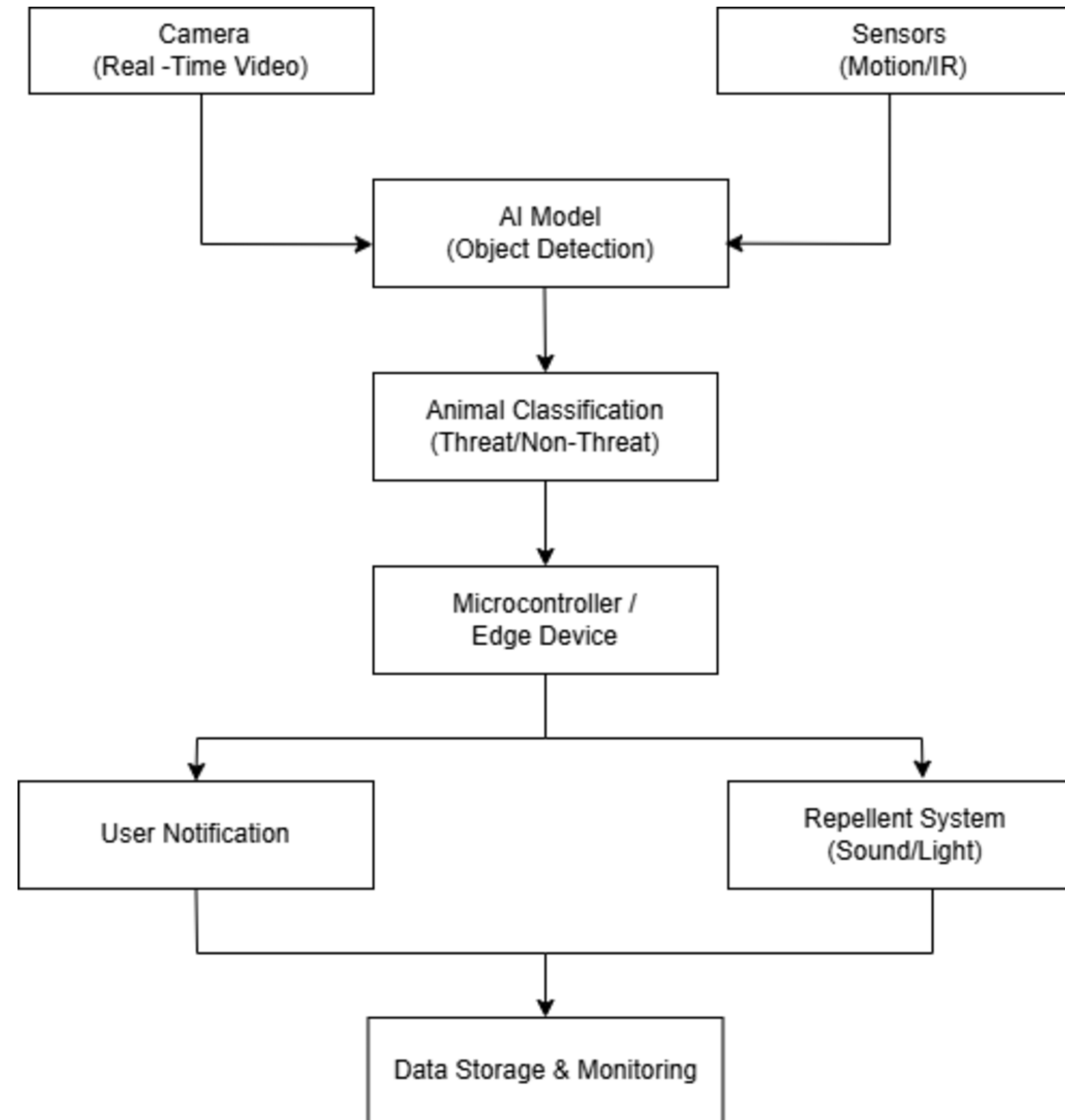
- Recent advancements in wildlife monitoring highlight the effectiveness of **integrating IoT, AI, and deep learning technologies**.
- Studies have shown that **YOLO-based models**, such as YOLOv5 and YOLOv7, achieve high accuracy in detecting animal intrusions, with mean average precision (mAP) reaching up to 94%.
- **IoT-based systems** using ultrasonic sensors and microcontrollers have proven scalable and efficient for real-time monitoring in agricultural and forest areas.
- **Deep learning techniques**, including depth-wise separable convolution layers, have demonstrated exceptional precision, with accuracy rates as high as 99.6% in wild animal detection.
- These approaches emphasize the potential of combining IoT, AI, and advanced machine learning for creating **scalable, real-time, and precise** wildlife monitoring systems.
- The WildEcoWatch project leverages these insights to develop an intelligent and efficient solution for wildlife conservation.

- The **WildEcoWatch** project integrates IoT and AI for real-time wildlife surveillance.
- The system begins by initializing **PIR sensors** to detect motion, activating **high-definition cameras** to capture video when motion is detected.
- Video data is processed on a **Raspberry Pi** using the **YOLOv8n** model implemented in **Python** and **OpenCV** to identify threats.
- Detected intrusions trigger real-time alerts via **MailJet** (email/SMS) and activate deterrents like **ultrasonic buzzers** and **LEDs**.
- Data is stored locally by updating the **excel sheet** or in the cloud for analysis.
- The system is to be tested for accuracy, deployed in pilot areas, and scaled up for broader use.
- Future enhancements include drone integration and advanced AI for behavioral analysis, ensuring a scalable and sustainable solution for wildlife conservation.

Flowchart



Block Diagram



Software & Hardware Details

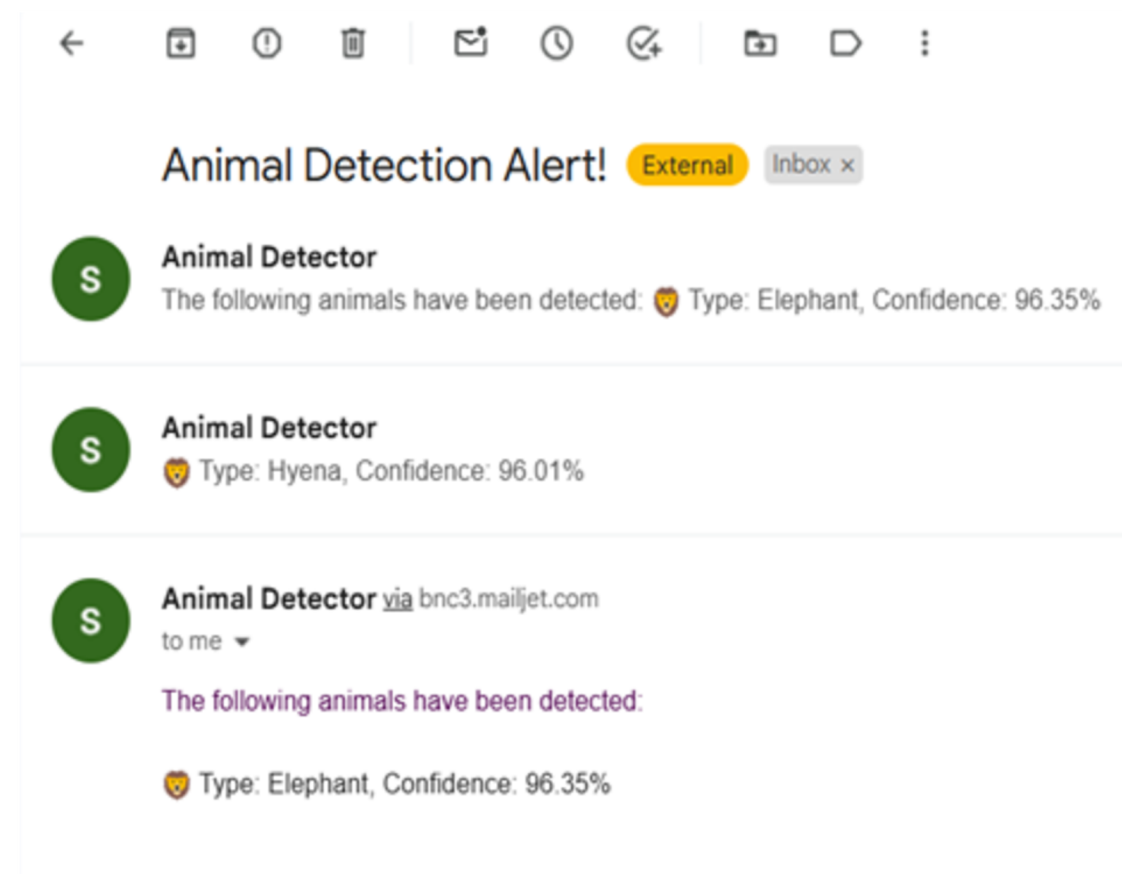
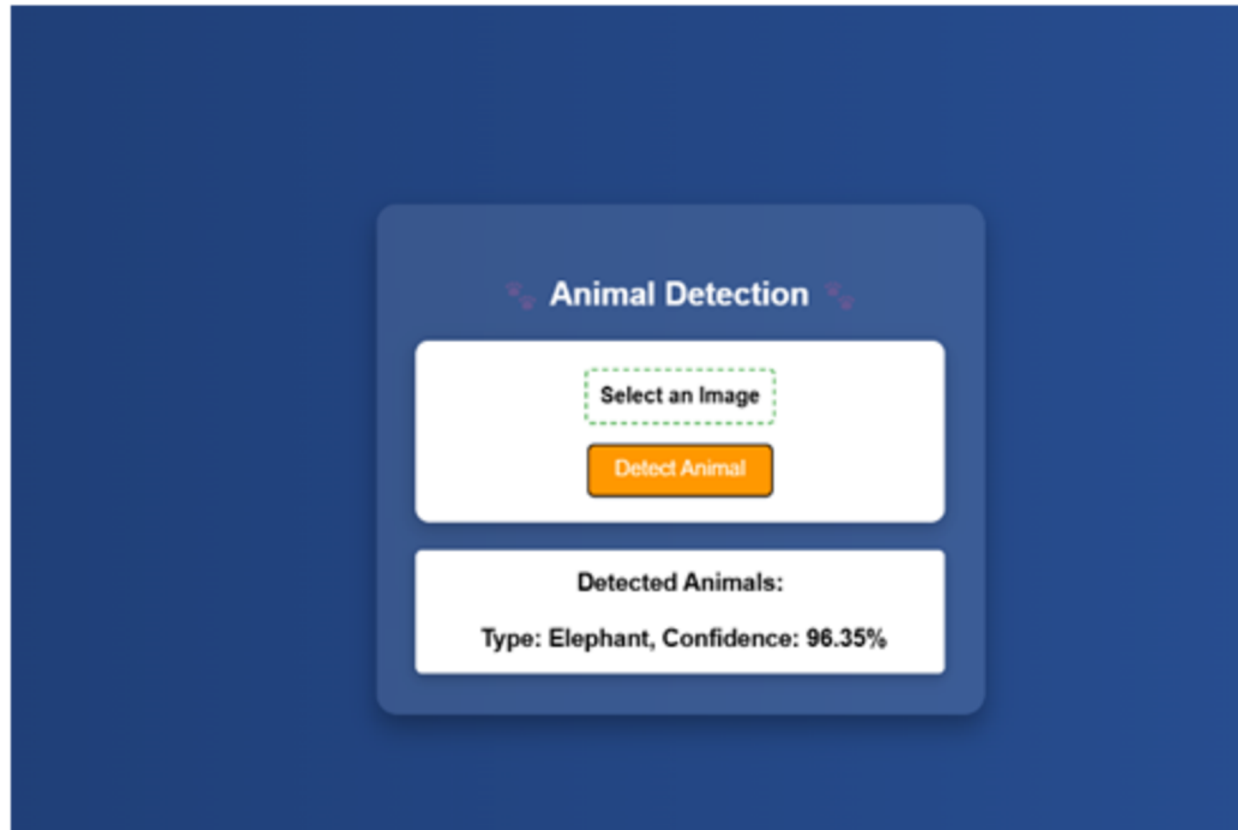
Software Details

- AI Model: Pre-trained object detection model YOLO V8
- Programming Language: Python
- Edge Device Frameworks: OpenCV
- Data Storage: Local or cloud-based storage solutions
- Notification Service: MailJet

Hardware Details

- Camera: High-definition night-vision-enabled camera
- Sensors: Motion detection sensors, PIR (passive infrared) sensor
- Microcontroller: Raspberry Pi
- Repellent Components: Sound generators Buzzer, LED flashlights

Results



Conclusion

- The WildEcoWatch project employs IoT and AI technologies to provide effective real-time monitoring of wildlife intrusions.
- This system enhances the ability of forest rangers and conservationists to respond quickly to potential threats.
- It operates sustainably in remote areas by utilizing low-power technology, minimizing infrastructure requirements.
- The project plays a crucial role in protecting endangered species and maintaining ecological balance through its monitoring capabilities.
- Ongoing collaboration among technologists, conservationists, and policymakers is essential to refine and expand these innovative wildlife management solutions.

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THANK YOU